Do Perturbation Models Compose Under Their Own Output?

Midterm Report — GenAI in Biomedicine, CMU, Fall 2026

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Abstract

Perturbation models predict a cell's expression after a genetic perturbation, and their output lives in the same space as their input. We ask whether feeding a prediction back in, $f(f(x, A), B)$, reproduces the measured double perturbation $A{+}B$, which a virtual-cell "world model" would require. On Norman et al. (2019), with all singles in training and all 131 doubles held out, the additive baseline (control + effect of A + effect of B) reaches Pearson δ 0.873. GEARS, trained only on singles, reaches 0.673 and has more than twice the additive error on the top-20 genes (0.181 vs. 0.082). A split-half design shows that non-additivity is a reproducible property of each pair (Spearman 0.94 across disjoint cells), and additive error rises fivefold from the least to the most non-additive tertile. Finally, GEARS's predicted shift does not depend on its input cell, so its rollout is additive by construction; scGPT is the model where composition remains an open question.

1 Introduction

Perturbation response models such as GEARS [3], scGPT [4] and State [8] predict how a cell's expression changes after a genetic perturbation. A control cell goes in; a predicted perturbed profile comes out. Because input and output share one expression space, a prediction can in principle be fed back in and perturbed again: predict(predict(x, A), B).

If a model has learned a reusable representation of cell state, that rollout should approximate the measured double perturbation $A{+}B$. If it is only a one-step map from control cells, it should not. As far as we know, no published work has tested which is the case.

The question matters because of where the field is heading. Proposals for virtual-cell world models [5] envision simulating cells under sequences of interventions. That needs a model whose output is a valid input. We do not build such a model; we test whether current models meet this precondition on the one dataset where it can be checked against measured combinations.

The starting point is sobering. Ahlmann-Eltze & Huber [2] showed that on Norman et al.'s [1] doubles, deep models do not beat simple addition of the two single effects. They evaluated only the joint prediction. We ask the complementary question: do models compose under their own output? Our test trains on single perturbations only, holds out every double, and compares three predictions per pair: the model's joint prediction of $A{+}B$, its sequential rollout A then B, and simple baselines, mainly addition.

Changes since the proposal, in response to reviewers.

1. **Two rollout arms (point 1).** Norman's doubles keep guide A active throughout, which neither rollout reproduces exactly. We report both *carried* (the second step gets label B only) and *restated* (it gets $A{+}B$).

2. **Round-trip gate (point 2).** No model enters the comparison until it passes a full input → output → input check on real and generated inputs.
3. **Split-half cells (point 3).** Each condition's cells are split into two random halves: one only for measuring non-additivity, the other for everything else.
4. **Two GEARS training regimes (point 4).** GEARS is trained on its own standard split as a diagnostic and on our strict singles-only split for the experiment.

We also narrowed scope to fit a 4.5-week calendar. The ComboSciPlex and Replogle experiment is cut. Geneformer is excluded because it outputs an embedding shift with no expression decoder. State and CPA are out for time.

2 Related Work

The closest prior work shows that simple addition beats deep models on Norman doubles; ours differs by testing rollout, using a singles-only split, and controlling for measurement noise.

Predicting doubles from singles. Norman et al. [1] measured 131 pairwise CRISPRa perturbations and all their singles in K562 cells. Most pairs combine roughly additively, with a minority showing synergy, buffering or new behaviour. The dataset is the standard benchmark for combinatorial prediction.

Deep perturbation models. GEARS [3] encodes each perturbed gene through a gene-ontology graph and predicts the perturbed expression of a control cell. scGPT [4] fine-tunes a pretrained transformer for the same task. CPA [6] and scGen [7] represent perturbations as vectors in a learned latent space. State [8] is a recent large transition model. All are trained and evaluated as one-step maps, control → perturbed.

Simple baselines beat deep models. Ahlmann-Eltze & Huber [2] showed that adding the two single shifts beats GEARS, scGPT and other foundation models on Norman doubles. Related benchmarks [9, 10, 11] found that much apparent accuracy comes from shifts shared by all perturbations. Table 1 summarises how our study differs.

Table 1: How this work differs from prior benchmarks.

	Prior benchmarks	This work
What is scored	Joint prediction $f(x, A{+}B)$ only	Joint and rollout $f(f(x, A), B)$, both arms
Training split	GEARS default; some doubles in training	All singles train, all 131 doubles test
Noise control	Strata and scores from the same cells	Strata from cell half 1, scores from half 2

Latent-space composition. scGen and CPA add perturbation vectors inside a latent space, the closest prior idea to rollout. There, composition is an assumption built into the architecture, not a property tested on measured doubles. It also happens in latent space rather than by re-entering the model through its expression input.

Virtual cells and world models. Xing & Song [5] argue for virtual cells that can be rolled forward under sequences of interventions. Such a model needs its outputs to be valid inputs, which is the precondition we test.

3 Methodology

We train on single perturbations only, score every held-out double on cells never used to select it, and compare models with the additive baseline pair by pair.

Research questions.

- **RQ1, joint.** Trained on singles only, does a model predict held-out doubles better than addition, and is any advantage concentrated in measurably non-additive pairs?
- **RQ2, composition.** Does the rollout $f(f(x, A), B)$ approach the measured double as well as the joint prediction?
- **RQ3, diagnosis.** When rollout fails, is it because the model cannot accept non-control input, or because its own predictions drift away from real cells?

Data. Norman et al. [1] CRISPRa Perturb-seq in K562 cells, GEARS-processed: 91,205 cells × 5,045 genes, 105 singles and 131 doubles. Every double has both of its singles measured.

Strict split. All singles and control are training conditions; all 131 doubles are test. GEARS's default split instead trains on some doubles. Under ours, a model and the additive baseline see the same information. For GEARS, 10% of the singles not involved in any double are held out for validation.

Split-half design. Each condition's cells are split at random into two halves, repeated over 5 seeds. Half 1 is used for measurement only: the non-additivity score η, and the top-20 differentially expressed genes (DEGs) used for scoring. Half 2 is used for everything else: inputs to baselines and models, and the targets they are scored against. The noise that puts a pair in a "highly non-additive" group therefore never appears in its scoring target. With Δ the pseudobulk shift from control in half h,

$$r_h = \Delta_{AB} - \Delta_A - \Delta_B, \qquad \eta_h = \|r_h\|. \tag{1}$$

Because the halves have independent noise, $r_1 \cdot r_2$ is an unbiased estimate of the true squared residual, which tells us how much of η is signal. Pairs are grouped into η tertiles using half 1.

Baselines. All are built from half-2 pseudobulks of training conditions (Table 2). Low-rank additive is the singles-only analogue of Ahlmann-Eltze & Huber's linear baseline.

Models. Each model sits behind one interface, $\text{predict}(X, \text{pert}) \to X$, which takes cells and returns cells in the same gene space. GEARS's own `predict()` samples control cells internally, so our adapter builds its input graph from the expression we pass. Predictions are anchored to the pseudobulk of their input condition, so models are not penalised for sampling noise in the ∼300 input cells. For each held-out pair we compute:

- **joint:** $f(x_0, \{A, B\})$

Table 2: Baselines and their prediction for A+B.

Baseline	Prediction
No change	control
Train mean	control + average single shift
Additive	control + $\Delta_A + \Delta_B$
Low-rank additive	control + rank-10 SVD-denoised $\Delta_A + \Delta_B$

- **carried rollout:** $f(f(x_0, A), B)$
- **restated rollout:** $f(f(x_0, A), \{A, B\})$
- **measured-input rollout:** $f(x_A, B)$, starting from real A-perturbed cells, which separates a failure to compose from a failure on generated input (RQ3)
- **both orders,** A then B and B then A

For GEARS on its standard split, doubles seen in training are excluded from scoring.

Round-trip gate. Before joining the comparison, a model must pass a check on singles. Real A-perturbed cells passed through with no new perturbation should come back nearly unchanged ($r > 0.95$), and so should the model's own predicted A state. Failure on real input means composition cannot be tested for that model; failure only on generated input means distribution shift.

Consistency checks. Order dependence ($A \to B$ vs. $B \to A$) and saturation (applying A twice) are reported as computational checks only. Norman measures joint perturbations and cannot validate either biologically.

Metrics and inference. MSE on the top-20 DEGs, and Pearson correlation of predicted vs. observed shift from control (Pearson δ), over all genes and over the top-20 DEGs. Per-pair results are averaged over the 5 seeds. All 95% intervals come from 2,000 bootstrap resamples over pairs, never cells, because the pair is the unit of inference.

4 Experiments (Preliminary Results)

Addition is the strongest predictor of held-out doubles, and GEARS trained on singles has more than twice its error. Setup: 131 held-out doubles, 5 random half-splits, per-pair results averaged over splits, 95% bootstrap intervals over pairs. GEARS used seed 1 and 20 epochs for each regime.

4.1 Baselines and GEARS joint prediction

Addition wins on every metric (Table 3, Figure 1). Denoising the singles makes it worse: low-rank addition beats plain addition on only 23 of 131 pairs.

GEARS is scored on fewer pairs for two reasons. Three doubles contain a gene outside GEARS's gene-ontology graph (IER5L+LYL1, RHOXF2BB+SET, RHOXF2BB+ZBTB25). On its standard split, GEARS also trains on 57 doubles, which are excluded. On the same pairs, addition scores 0.082 (128 pairs) and 0.085 (71 pairs), so the gap is not a pair-set artefact.

On the strict split, GEARS's MSE exceeds addition's by 0.100 [0.077, 0.122], and GEARS is better on only 14% of pairs. On the standard split the gap is 0.148 [0.106, 0.187], with GEARS

Table 3: Held-out doubles, mean [95% CI]. Lower MSE and higher Pearson δ are better. No-change predicts zero shift, so its correlation is undefined.

Predictor	Pairs	MSE, top-20 DEGs	Pearson δ, all genes	Pearson δ, top-20 DEGs
Additive	131	**0.081** [0.067, 0.096]	**0.873** [0.857, 0.886]	**0.959** [0.952, 0.964]
Low-rank additive	131	0.140 [0.123, 0.158]	0.789 [0.766, 0.810]	0.870 [0.847, 0.891]
GEARS, strict split	128	0.181 [0.161, 0.204]	0.673 [0.645, 0.700]	0.862 [0.833, 0.888]
GEARS, standard split	71	0.232 [0.194, 0.271]	0.605 [0.558, 0.653]	0.798 [0.740, 0.851]
Train mean	131	0.575 [0.494, 0.656]	0.533 [0.488, 0.575]	0.644 [0.582, 0.700]
No change	131	0.734 [0.643, 0.824]	undefined	undefined

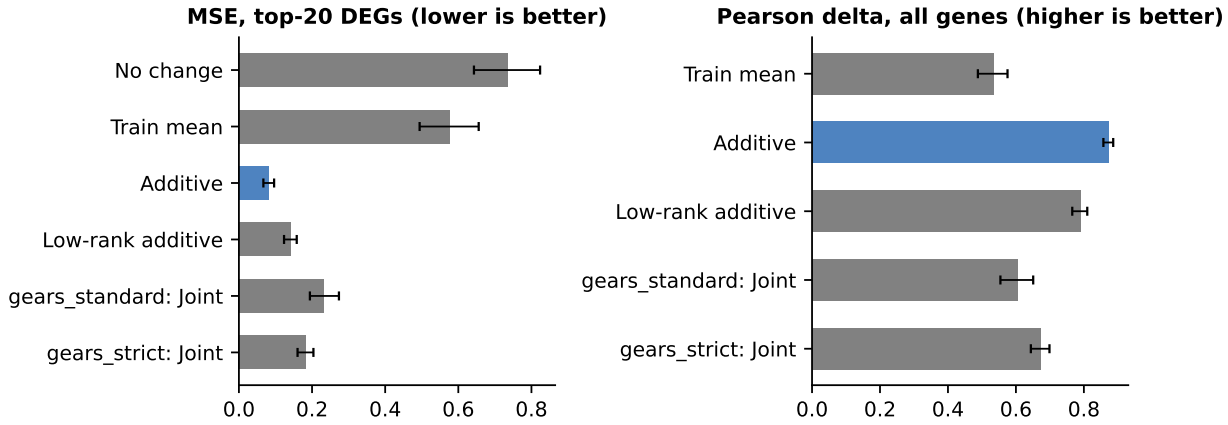


Figure 1: Baselines and GEARS joint prediction on held-out doubles, with 95% bootstrap intervals over pairs.

better on 13%. GEARS beats only the train-mean and no-change baselines. The two GEARS rows are not directly comparable, because the standard set includes pairs with a single never seen in training.

4.2 Non-additivity is a reproducible property of each pair

η measured on two disjoint cell halves agrees with Spearman 0.944 (per seed 0.932 to 0.955; Figure 2). The cross-half estimate of true η^2 is positive for 99.5% of pairs and accounts for 61% of measured η^2. η is therefore mostly signal, with a noise share of about 40%.

4.3 Additive error rises with non-additivity

With η taken from half 1 and error from half 2, η and additive MSE correlate with Spearman 0.82 across pairs (Figure 3). Additive error rises about fivefold from the low to the high tertile; GEARS stays above it throughout, and the gap narrows only where interactions are strongest (Table 4). The most non-additive pairs are dominated by the CEBP family (CEBPA+CEBPB, CEBPA+CEBPE, CEBPE+SET, CEBPA+KLF1), consistent with the redundant differentiation programmes Norman et al. described.

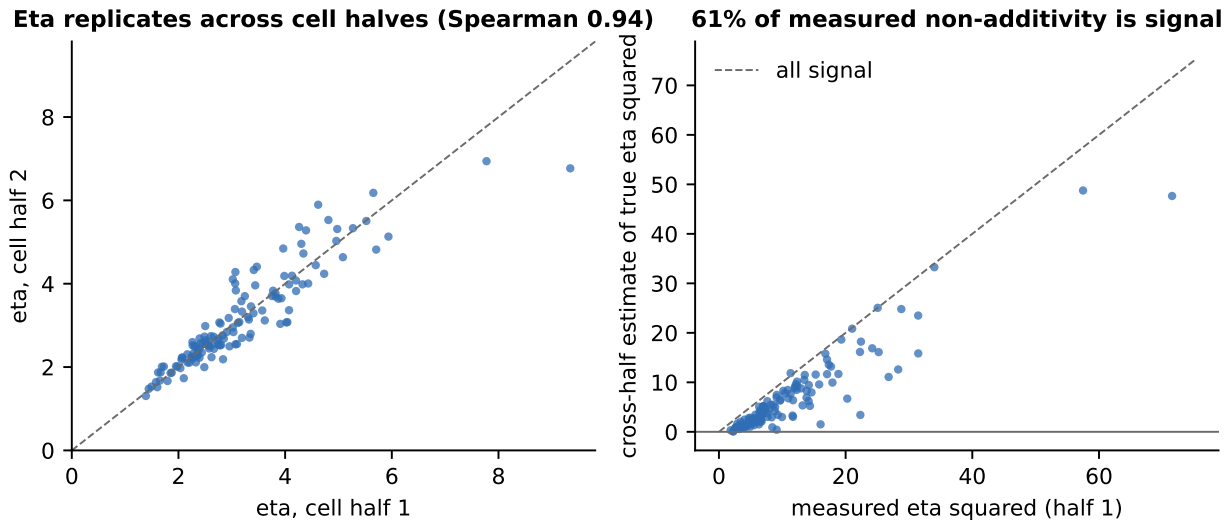


Figure 2: Left: η measured on two disjoint cell halves. Right: the cross-half estimate of true η^2 against measured η^2; the dashed line is "all signal".

Table 4: MSE on the top-20 DEGs by η tertile, strict split, same 128 doubles.

η tertile	Additive	GEARS, strict
Low	0.028	0.140
Mid	0.064	0.210
High	0.153	0.194

4.4 Round-trip gate

Strict GEARS passes on the 5 singles with the most cells (KLF1, BAK1, CEBPE, ETS2, UBASH3B). Real A-perturbed cells passed through with no new perturbation come back with $r = 0.999$ on expression and 0.80 to 0.99 on shifts from control. Its own predicted states behave the same.

The pass is structural, not learned. Applying A twice moves the cell exactly as far as applying it once (saturation ratio 1.0 for all five genes). The GEARS source explains why: input expression enters only as a final residual, out $= x + g(\text{pert})$, and g ignores the input cell. GEARS's prediction is therefore independent of the state it starts from.

5 Analysis and Ablations

The midterm results already answer RQ1 for GEARS (it does not beat addition) and reduce RQ2 for GEARS to a comparison of two additive constructions.

η is a sound way to group pairs. The worry was that "highly non-additive" pairs might just be the noisiest ones, making their high error a regression-to-the-mean artefact. The split-half design rules this out: η replicates across disjoint cells (Spearman 0.94), and every error is scored on the half not used to define the groups. The high-η tertile is where a model with real interaction knowledge has room to beat addition.

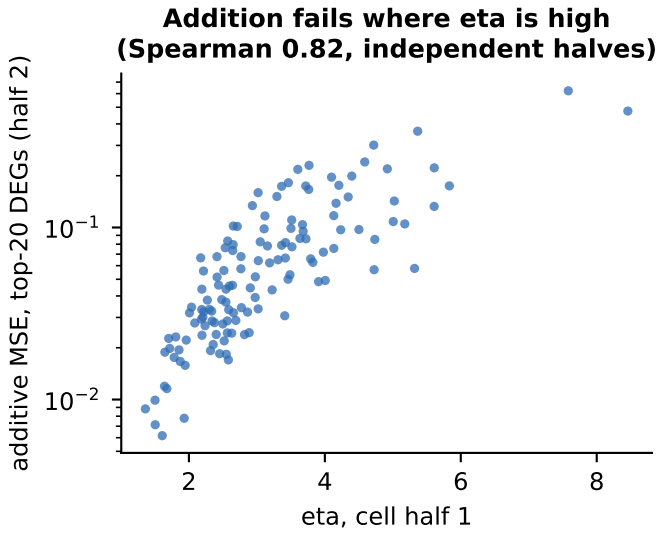


Figure 3: Additive error (half 2) against η (half 1), one point per held-out double.

Addition is a hard bar, and the room above it is concentrated. Most of addition's error sits in the top η tertile, five times the low tertile's. An overall average therefore mostly reflects easy, near-additive pairs. The rollout analysis after the midterm is planned as advantage over addition as a function of η, not as one average.

Ablation: smoothing the singles does not help. Low-rank addition is worse than plain addition on 82% of pairs. With pseudobulks from roughly 100 to 300 cells, the single effects are already precise, so the rank-10 projection removes more real signal than noise. A singles-only model must know something about how effects interact to beat addition.

GEARS has not learned interactions, and it cannot carry state. On the strict split, GEARS's error exceeds addition's in every tertile. Its gap narrows in the high tertile (0.194 vs. 0.153) but does not reverse. Its predicted shift also ignores the input cell, so its rollouts are fixed by construction:

$$\text{carried: } x + g(A) + g(B), \qquad \text{restated: } x + g(A) + g(A, B). \tag{2}$$

Order cannot matter, and applying A twice simply doubles its shift. For GEARS, the remaining empirical question is whether its own singles, $g(A) + g(B)$, land closer to the measured double than its joint $g(A, B)$. scGPT, whose output does depend on the input, is where composition is a genuinely open question.

Caveat: η partly tracks effect size. η correlates with the size of the double's shift (Spearman 0.54), and shift size alone predicts additive error about as well (0.58). η relative to the shift barely predicts it (0.15), and the scale-free Pearson δ hardly falls across tertiles (0.880 to 0.862). Part of the trend is "bigger perturbations have bigger absolute errors". For the final report we will compare models with addition within tertiles, which cancels scale, and check conclusions with relative η or effect size as a covariate.

6 Limitations

The midterm results rest on one dataset, one GEARS seed and joint predictions only.

- **One dataset.** Norman is the only public set with doubles and all their singles, so we cannot test generality on another cell type or perturbation modality.
- **Biology of rollout.** Norman's guides act together for the whole experiment. Neither rollout arm matches that exactly, and order and saturation cannot be validated biologically.
- **One GEARS run.** Each regime was trained once (seed 1, 20 epochs, default hyperparameters). Seed-to-seed variation and tuning are untested.
- **Unpredictable pairs.** GEARS cannot perturb RHOXF2BB, KIAA1804 or IER5L, so 3 of 131 doubles are missing from its scores.
- **Effect-size confound.** η partly measures the size of the perturbation, not only its non-additivity (Section 5).
- **Rollout not yet run.** The midterm reports joint predictions only; the composition results are still to come.

7 Timeline for Remaining Work

The full rollout experiment runs next; by November 15 we decide whether scGPT passes the round-trip gate or the study proceeds with GEARS alone.

Table 5: Remaining milestones.

Date	Milestone
Nov 6	Midterm report (this document)
Nov 15	Decide: keep scGPT, or GEARS only
Nov 22	Final-report figures drafted
Dec 3	Final report and presentation

Before the November 15 decision.

1. Full rollout on strict GEARS: both arms, both orders, and the measured-input control (Shreyan).
2. Standard-split GEARS rollout as the diagnostic comparison (Shreyan).
3. scGPT adapter, then the round-trip gate; joint and rollout only if it passes (Shreyan).
4. Rollout analysis: advantage over addition vs. η with slope and CI, order dependence vs. η, arm comparison (Datta).

After it.

1. Stability: saturation ratio and drift from real cells over 1 to 10 steps (Datta).
2. Effect-size checks from Section 5: within-tertile comparisons and relative η (Datta).
3. Draft final-report figures by November 22; final report and presentation by December 3 (both).

Cut order if time runs short: the discriminator-based drift metric, then five split seeds down to three, then scGPT, then the stability experiment.

References

[1] T. M. Norman et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 2019. https://doi.org/10.1126/science.aax4438

[2] C. Ahlmann-Eltze and W. Huber. Deep learning-based predictions of gene perturbation effects do not yet outperform simple linear baselines. *Nature Methods*, 2025. https://doi.org/10.1038/s41592-025-02772-6

[3] Y. Roohani, K. Huang, and J. Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology*, 2024. https://doi.org/10.1038/s41587-023-01905-6

[4] H. Cui et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 2024. https://doi.org/10.1038/s41592-024-02201-0

[5] Xing and Song. Virtual cell world model. *Cell*, 2026.

[6] M. Lotfollahi et al. Predicting cellular responses to complex perturbations in high-throughput screens. *Molecular Systems Biology*, 2023.

[7] M. Lotfollahi, F. A. Wolf, and F. J. Theis. scGen predicts single-cell perturbation responses. *Nature Methods*, 2019.

[8] Arc Institute. State: a transition model for cellular perturbation response, 2025.

[9] A. Wenteler et al. PertEval-scFM: benchmarking single-cell foundation models for perturbation effect prediction.

[10] G. Csendes et al. Benchmarking foundation cell models for post-perturbation RNA-seq prediction.

[11] R. Viñas Torné et al. Systema: systematic variation in perturbation response prediction.